



Tribhuvan University
Faculty of Humanities and Social Sciences

ONLINE JOB PORTAL

A PROJECT REPORT

Submitted to
Department of Computer Applications
Jana Bhawana Campus

In partial fulfillment of the requirements for the Bachelor's in Computer Application

Submitted by
Rohit Shrestha (25302095)
Nikhil Tamang (25302090)

03-2026

Under the Supervision of
Mr. Niraj Acharya



Tribhuvan University

Faculty of Humanities and Social Sciences

Jana Bhawana Campus

Supervisor's Recommendation

I hereby recommend that this project prepared under supervision by NIKHIL TAMANG and ROHIT SHRESTHA entitled “**Online Job Portal**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

.....

SIGNATURE

Mr. Niraj Acharya

SUPERVISOR

BCA Department

Jana Bhawana Campus

Godawari – 11, Chapagaun



Tribhuvan University

Faculty of Humanities and Social Sciences

Jana Bhawana Campus

LETTER OF APPROVAL

This is to certify that this project prepared by NIKHIL TAMANG and ROHIT SHRESTHA entitled **“Online Job Portal”** in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion, it is satisfactory in the scope and quality as a project for the required degree.

SIGNATURE of Supervisor Name – Mr. Niraj Acharya Department of BCA Godawari – 11, Chapagaun	SIGNATURE of HOD/ Coordinator Name – Mr. Sanam Dongol Department of BCA Godawari – 11, Chapagaun
SIGNATURE of Internal Examiner Name: Date:	SIGNATURE of External Examiner Name: Date:

ABSTRACT

The growing demand for a digital platform to connect job seekers and employers has led to the development of an Online Job Portal. This project aims to create a basic yet effective web-based platform that allows job seekers to browse, search, and apply for job opportunities, while employers can post job vacancies and manage applications efficiently. The system is designed to bridge the gap between job providers and job seekers by offering a centralized and organized environment for recruitment.

The Online Job Portal includes essential features such as user registration and login, job listings, advanced search and filter options, and the ability for users to upload and manage their resumes. Job seekers can explore various job categories, view detailed job descriptions, and apply directly through the platform. On the other hand, employers can create job postings, review submitted applications, and shortlist candidates based on their requirements.

In addition to these core functionalities, the system focuses on providing a user-friendly interface that is easy to navigate, ensuring accessibility for users with different levels of technical knowledge. The platform is developed using modern web technologies to ensure responsiveness, reliability, and efficiency.

The main objective of this project is to simplify the job search and recruitment process by reducing manual efforts and improving communication between employers and candidates. By developing this platform, we aim to contribute to the growing digital landscape of job recruitment, making the hiring process faster, more transparent, and more convenient for both job seekers and employers.

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to everyone who supported us throughout the development of the Online Job Portal project.

First and foremost, we extend our sincere thanks to our respected teacher and project supervisor, Mr. Niraj Acharya, whose guidance, valuable suggestions, and encouragement were instrumental in the successful completion of this project. His continuous support motivated us to refine our ideas and gain deeper insights into web development and database management.

We are also grateful to Jana Bhawana Campus and the Department of Computer Applications for providing us with the necessary resources, a conducive learning environment, and the opportunity to apply our academic knowledge in a practical project that addresses real-world challenges in online job recruitment.

Furthermore, we would like to thank our friends and classmates for their cooperation, constructive feedback, and motivation during the development process. Their inputs helped us identify shortcomings and improve the usability of the system for both employers and job seekers.

Finally, we are thankful to everyone who directly or indirectly contributed to the successful completion of this project. This project has been a valuable learning experience in developing user-centric web applications with multi-role functionality.

Rohit Shrestha

Nikhil Tamang

TABLE OF CONTENTS

SUPERVISOR'S RECOMMENDATION	i
LETTER OF APPROVAL	ii
ABSTRACT	iii
ACKNOWLEDGEMENT	iv
LIST OF ABBREVIATIONS.....	vii
LIST OF FIGURES.....	viii
LIST OF TABLES.....	ix
CHAPTER 1: INTRODUCTION	1
1.1 Introduction.....	1
1.2 Problem Statement	1
1.3 Objectives.....	2
1.4 Scope and Limitation	2
1.4.1 Scope of the System.....	2
1.4.2 Limitations of the System	2
1.5 Report Organization.....	2
CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW	4
2.1 Background Study.....	4
2.2 Literature Review.....	5
CHAPTER 3: SYSTEM ANALYSIS AND DESIGN	6
3.1 System Analysis	6
3.1.1 Methodology	6
3.1.2 Requirement Analysis	7
3.1.3 Feasibility Analysis.....	9
3.1.4 Data Modeling (ER-Diagram)	10
3.1.5 Process Modeling (DFD)	11
3.2 System Design.....	12

3.2.1 Architectural Design	12
3.2.2 Database Schema Design	13
3.2.3 Interface Design	14
CHAPTER 4: IMPLEMENTATION AND TESTING	15
4.1 Implementation	15
4.1.1 Tools Used.....	15
4.1.2 Implementation Details	16
4.2 Testing	17
CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS	23
5.1 Lesson Learnt / Outcome	23
5.2 Conclusion	23
5.3 Future Recommendations	24
REFERENCES.....	25
APPENDICES.....	26

LIST OF ABBREVIATIONS

Abbreviation	Full Form
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DB	Database
DBMS	Database Management System
HTML	Hypertext Mark-up Language
ID	Identification Number
JS	JavaScript
PHP	Hypertext Pre-processor
SQL	Structured Query Language
XAMPP	Cross-Platform Apache MySQL PHP Perl

LIST OF FIGURES

Figure 3.1 Iterative Waterfall Model.....	6
Figure 3.2 Use case diagram	8
Figure 3.3 Gantt Chart.....	9
Figure 3.4 E-R Diagram For Online Bookstore	10
Figure 3.5 Level 0 DFD For Online Bookstore	11
Figure 3.6 DFD Level 1 For Online Bookstore	11
Figure 3.7 Schema Diagram For Online Bookstore.....	13
Figure 3.8 Figma Design for user Dashboard	14
Figure 3.9 Figma Design for Admin Dashboard.....	16

LIST OF TABLES

Table 4.2.1.1 User Authentication and Registration.....	18
Table 4.2.1.2 Book, Cart, Wishlist, Order Management	19
Table 4.2.2.1 Test Cases for System Testing	20
Table 4.2.3.1 Integrated System Testing	22

Chapter 1: Introduction

1.1 Introduction

The Online Job Portal is a digital platform designed to connect job seekers with potential employers. It provides a centralized system where individuals looking for employment can create profiles, upload resumes, and explore various job opportunities across multiple industries and locations. At the same time, companies can post vacancies, manage applications, and identify suitable candidates efficiently. This system aims to address the growing demand for an organized and accessible medium for recruitment.

This platform serves as a bridge between job seekers and organizations, streamlining the entire recruitment process. By offering tools such as search, filtering options, and application tracking, the portal ensures that job seekers can find opportunities that match their skills and preferences, while employers can reach qualified candidates quickly. The portal's design emphasizes ease of use, making it accessible even for users with limited technical experience.

The primary objective of the Online Job Portal is to simplify the job search and hiring process, reducing manual effort and improving communication between employers and candidates. By developing this platform, the project contributes to the digital transformation of job recruitment, offering an efficient, reliable, and user-friendly solution that benefits both job seekers and employers.

1.2 Problem Statement

The current job search and recruitment process faces several challenges:

- **Limited Accessibility:** Job seekers often struggle to find suitable opportunities, especially in smaller towns, while employers may find it difficult to reach qualified candidates without a centralized platform.
- **Lack of Convenience:** Traditional recruitment methods, such as job fairs or offline applications, require significant time and effort, whereas an online portal provides instant access to job listings and applications.
- **No Personalization:** Existing systems rarely allow users to save preferences or track applications, limiting engagement and efficiency.
- **Administrative Inefficiency:** Managing job postings, applications, and employer accounts manually is time-consuming and prone to errors.

1.3 Objectives

The primary objectives of this project are:

- **Develop Online Platform:** To create a web-based portal where employers can post job vacancies and manage applications efficiently, while job seekers can create profiles, upload resumes, and search for relevant job opportunities.
- **Facilitate Efficient Job Matching:** To help job seekers find suitable jobs using advanced search and filtering options, and assist employers in shortlisting qualified candidates quickly.
- **Simplify Recruitment Process:** To streamline the application process for both job seekers and employers, making it easy to apply, review, and manage applications through a centralized platform.

1.4 Scope and Limitation

1.4.1 Scope of the System

The Online Job Portal includes the following functionalities:

- **User Management:** Job seekers can create accounts, manage profiles, and track applications, while employers can post jobs and manage applications.
- **Job Search & Application:** Job seekers can browse, search, and apply for jobs directly through the portal.
- **Employer Oversight:** Employers can post, edit, or delete job vacancies and shortlist candidates efficiently.
- **Centralized System:** All updates by employers are immediately reflected on the platform, keeping information accurate and up-to-date.

1.4.2 Limitations of the System

The Online Job Portal has the following limitations:

- **Restricted Features for Employers:** The system does not include advanced recruitment tools such as applicant scoring or interview scheduling.
- **Limited Candidate Interaction:** Job seekers and employers cannot communicate directly through the portal.
- **Limited Personalization:** The portal only allows users to track the status of their applications; it does not provide personalized job recommendations or alerts.

1.5 Report Organization

The project report is structured into the following five chapters:

- **Chapter 1:** Introduction – Provides an overview of the Online Job Portal, the challenges in traditional job search and recruitment, and the system's objectives, scope, and limitations.
- **Chapter 2:** Background Study and Literature Review – Explores the digital transformation in recruitment, the evolution of online job portals, and existing research on web-based job management systems.
- **Chapter 3:** System Analysis and Design – Details the iterative waterfall methodology, requirement analysis for the two user roles (Employer and Job Seeker), and technical modeling using Data Flow Diagrams (DFDs) and ER diagrams.
- **Chapter 4:** Implementation and Testing – Describes the use of web technologies (HTML, CSS, JavaScript, PHP, MySQL), module-specific logic (job posting, application management, and search functionality), and comprehensive test cases for system validation.
- **Chapter 5:** Conclusion and Future Recommendations – Summarizes project outcomes, highlights the benefits to employers and job seekers, and suggests future enhancements such as advanced personalization, messaging features, and analytics dashboards.

Chapter 2: Background Study and Literature Review

2.1 Background Study

The recruitment and hiring industry has undergone significant digital transformation, with online job portals emerging as essential platforms connecting job seekers and employers. Traditional recruitment methods, such as newspaper advertisements and manual applications, often face limitations including restricted reach, slower communication, and lack of accessibility, reducing efficiency for both parties.

Online job portals overcome these challenges by providing centralized systems where employers post job vacancies and seekers explore and apply for opportunities, improving accessibility, reducing hiring time, and streamlining recruitment.

Existing systems generally follow two approaches:

- Open Listing Model: all jobs visible without filtering.
- Preference-Based Filtering Model: users select areas of interest, and the system displays matching jobs.

The proposed system adopts a preference-based filtering approach, where job seekers choose preferred job categories during profile setup. Job listings are then displayed according to these preferences, improving relevance while keeping the implementation simple compared to complex recommendation systems.

Key technologies include PHP for server-side scripting, MySQL for database management, and HTML/CSS for frontend development. Secure authentication is ensured with password hashing and session management, and email notifications via PHPMailer keep users informed about application status.

A primary challenge is balancing functionality with simplicity, providing tools for administrators and employers while ensuring an intuitive experience for seekers. This system highlights the value of structured data management, role-based access control, and user-defined preference filtering for an effective online job portal.

2.2 Literature Review

Digital recruitment platforms have transformed the hiring process, streamlining interactions between job seekers and employers. According to Sundararajan [1], online job portals increase the efficiency of the recruitment cycle by enabling centralized job postings, automated application tracking, and role-based access for different users. Similarly, Kaur and others highlighted that preference-based filtering and personalized dashboards significantly improve job seeker engagement and reduce information overload, thereby increasing the likelihood of relevant applications [2].

Case studies of leading platforms demonstrate practical implementation of these principles. LinkedIn provides role-specific dashboards and job suggestions based on explicit user preferences, ensuring that users can quickly identify suitable opportunities [3]. Indeed.com emphasizes straightforward job categorization and keyword-based search, making the platform accessible while still supporting user-centric interaction [4]. These studies underline the importance of combining structured data management with personalized features to optimize user experience and system effectiveness.

The technical infrastructure plays a central role in enabling these features. PHP and MySQL are commonly employed for server-side processing and database management, supporting multi-role architectures where administrators, employers, and job seekers have distinct functionalities [5]. Additionally, integrating email notification systems (e.g., SMTP via PHPMailer) ensures that application status updates reach seekers promptly, reflecting real-world communication practices and improving platform reliability [6].

Overall, literature indicates that a combination of preference-based filtering, role-based dashboards, and robust backend technologies is essential for creating effective online job portals that balance usability, personalization, and administrative control.

Chapter 3: System Analysis and Design

3.1 System Analysis

3.1.1 Methodology

The SDLC model used for the development of the Online Job Portal system is the Iterative Waterfall Model. This model is a structured approach to software development that retains the sequential phases of the classical waterfall—requirements analysis, system design, implementation, testing, and maintenance—while introducing feedback loops. Unlike the traditional waterfall, the iterative version allows revisiting earlier phases when issues are identified, making it more flexible and practical for projects that require refinement during development.

The iterative waterfall methodology was chosen for this project because it provides a clear structure and organized development process while allowing flexibility to adapt to feedback. It is suitable for projects like the Online Job Portal, where requirements are mostly defined but occasional adjustments are needed to enhance usability, performance, and job matching accuracy. By combining structured phases with iterative feedback, this model ensured that the portal was developed efficiently, with opportunities to improve quality and user experience at each stage.

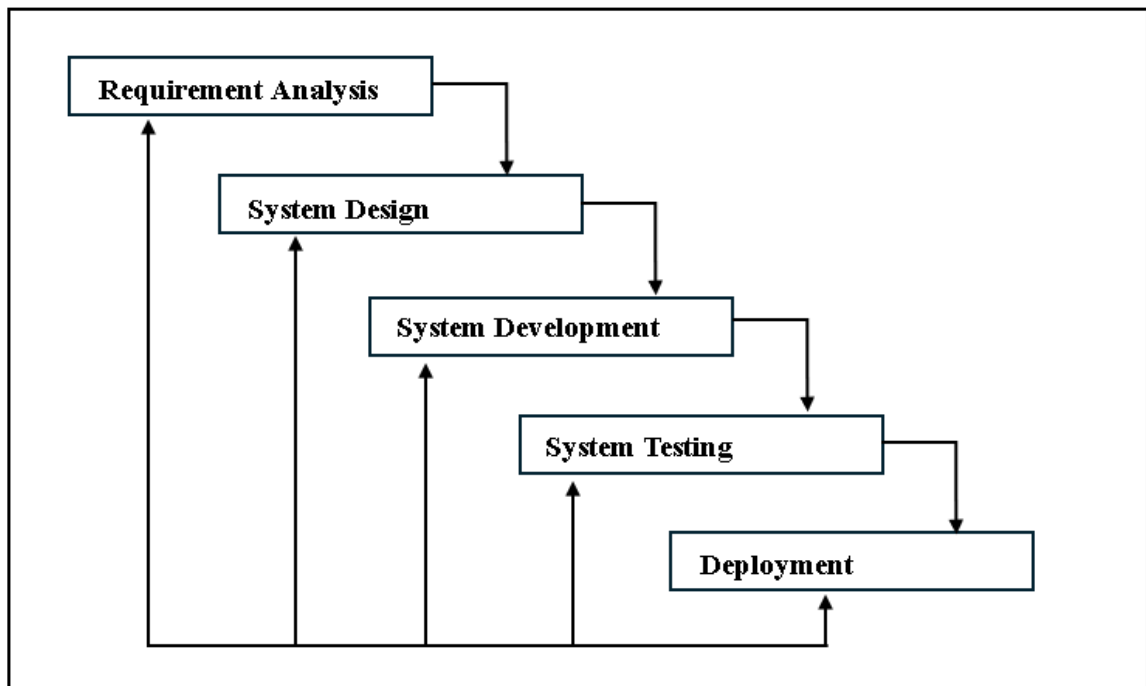


Figure 3.1: Iterative Waterfall Model of Online Job Portal

3.1.2 Requirement Analysis

Functional Requirements

a) Job Seeker Module:

- Secure Registration & Profile Setup: Create an account with personal and professional details, and upload a PDF resume.
- Job Search & Browsing: View all available job postings; search jobs by keyword (title, location, company).
- Job Application System: Apply for jobs with confirmation, check deadlines, and prevent duplicate applications.
- Application Status Tracking: View the status of applied jobs (pending, accepted, rejected) in the dashboard.

b) Employer Module:

- Secure Registration & Profile Setup: Create an account with company details and industry information.
- Job Posting & Management: Create, edit, and delete job postings with title, description, location, salary, deadline, and category.
- Applicants Review System: View all applicants for each job with resume, application date, and contact info.
- Application Status Updates: Accept or reject applicants; triggers email notifications to seekers via PHPMailer.

c) Admin Module:

- Centralized Dashboard: View total users, job postings, and overall system activity.
- User Management: View, edit, and delete any user account (seeker or employer).
- Job Management: View, edit, and delete any job posting; changes reflected in real-time on dashboards.
- Content Moderation: Ensure quality and relevance of all jobs and user profiles.

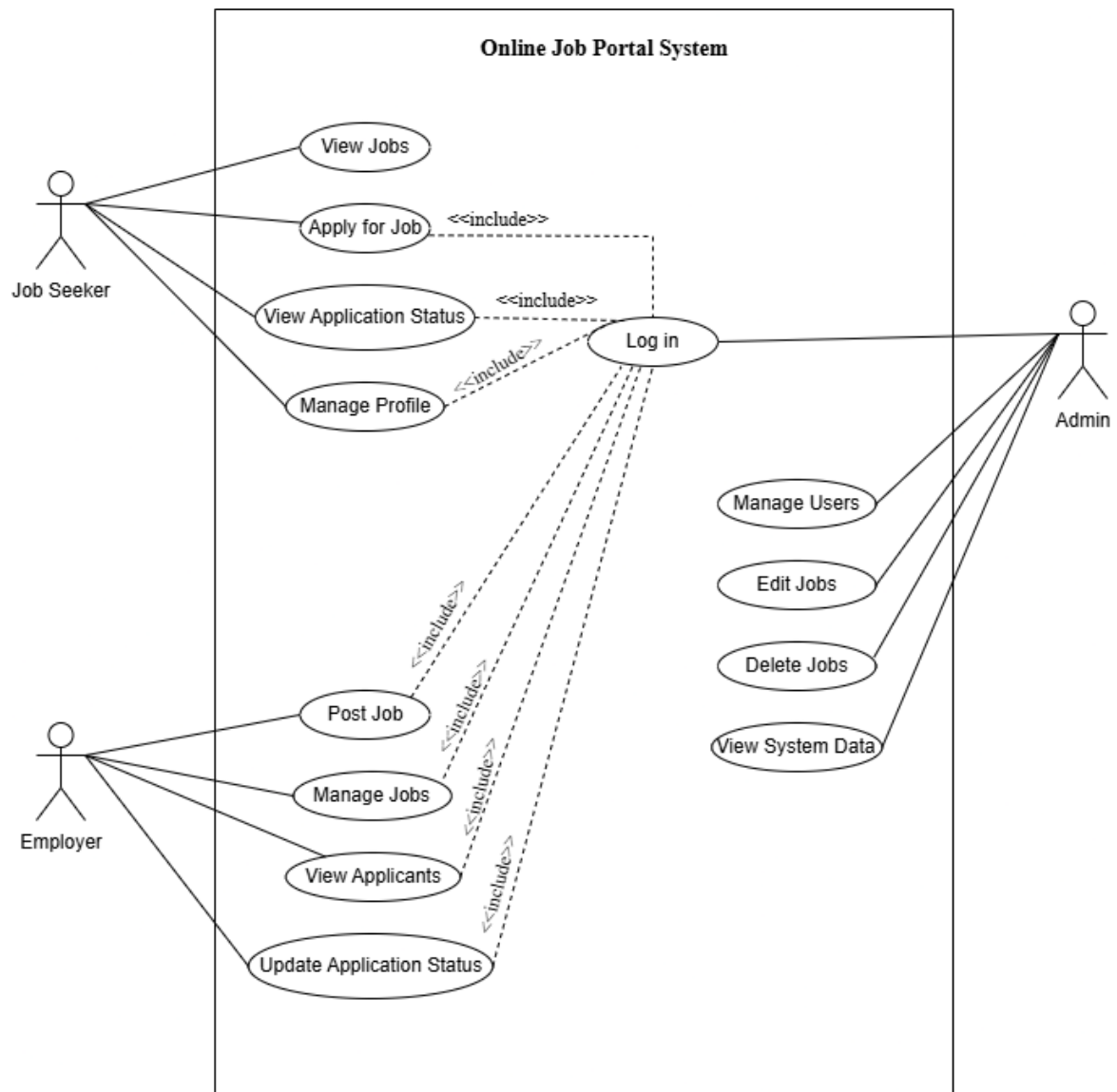


Figure 3.2 Use Case Diagram of Online Job Portal

Non-Functional Requirements

- **Performance:** The system should load job listings and detail pages within a few seconds to ensure smooth browsing and quick response time.
- **Security:** All sensitive user data, especially passwords, must be encrypted and protected using secure methods like hashing and prepared statements.
- **Usability:** The user interface (UI) must be responsive and easy to navigate. Dashboards should provide a clean layout for managing jobs, applications, and profiles.
- **Maintainability:** The codebase should follow a modular structure by separating logic, design, and database files, allowing easy updates and CRUD operations.

3.1.3 Feasibility Analysis

- i. **Technical Feasibility:** The system can be developed using commonly available technologies like HTML, CSS, JavaScript, PHP and MySQL. These tools are easy to use, open-source and compatible with most systems. Hence the project is technically feasible.
- ii. **Economic Feasibility:** The project does not require expensive hardware or licensed software. It uses free tools like XAMPP and VS Code, keeping the development cost minimal. Therefore, the project is economically feasible.
- iii. **Operational Feasibility:** The system is designed to be simple and user-friendly, allowing job seekers, employers, and admin to use it easily with basic computer knowledge. The admin can manage jobs and users efficiently without extensive training. Thus, the project is operationally feasible.
- iv. **Schedule Feasibility:** The project was planned to be completed within 12–14 weeks, including requirement analysis, system design, implementation, testing, and documentation. Most core features such as user registration, job posting, job application, and dashboards have been implemented on time, and testing is in progress. Therefore, the project is schedule feasible.

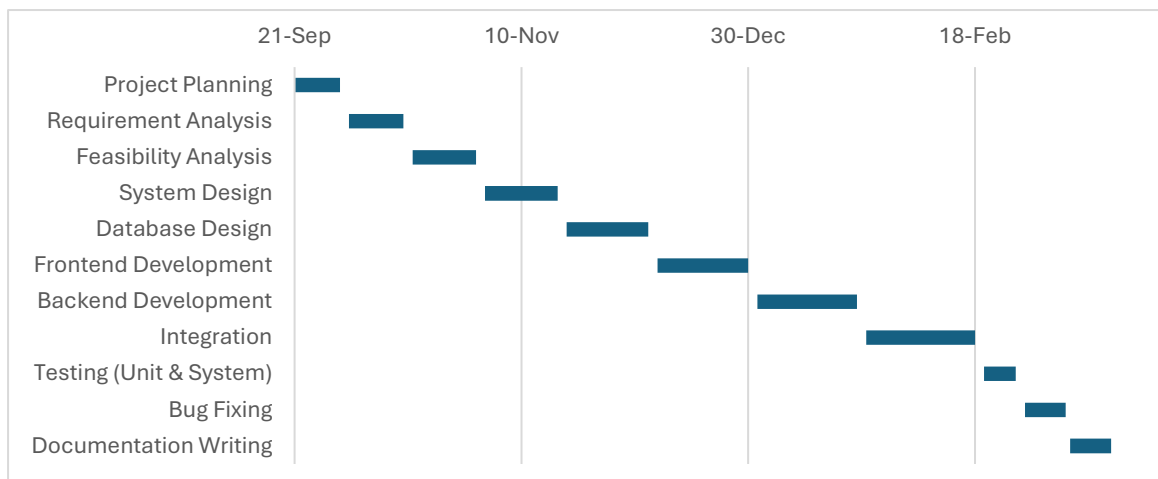


Figure 3.3: Gantt Chart of Online Job Portal

3.1.4 Data Modeling (ER-Diagram)

The Online Bookstore System is a web-based e-commerce platform designed to simplify book retailing by directly connecting readers with a centralized admin role. Developed using Core PHP and MySQL, the system follows a modular architecture for stability and easy maintenance. Two main user roles are supported: Admin and Buyer. The Admin manages book listings, inventory, pricing, and platform security, while Buyers can browse categories, add books to their cart, and complete purchases securely.

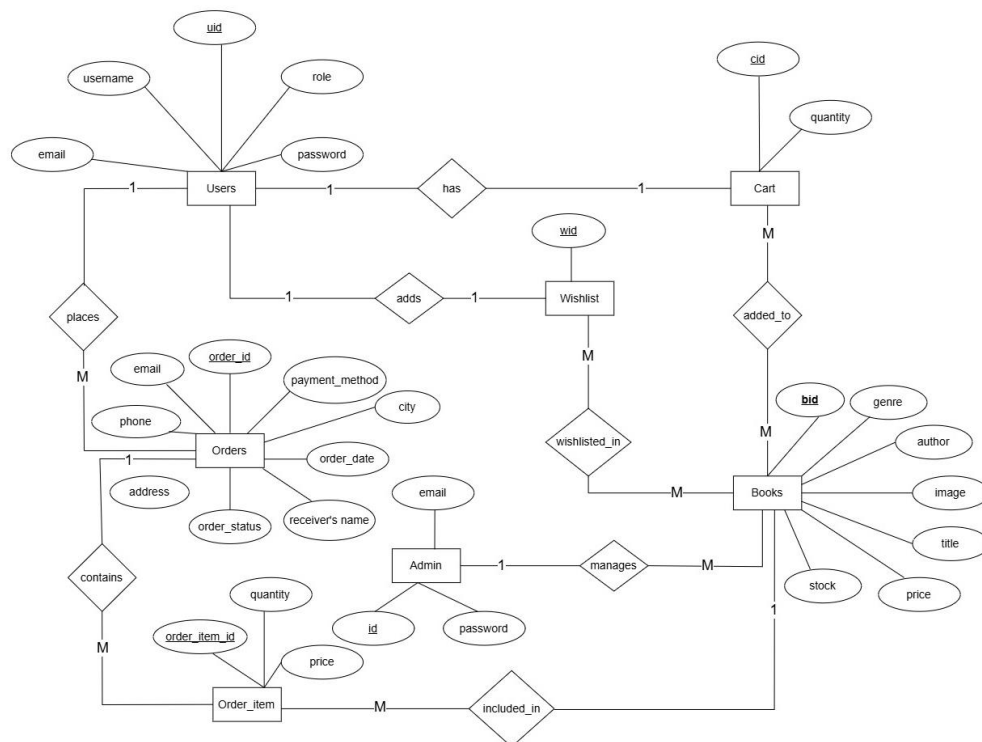


Figure 3.4: ER-Diagram of Online Bookstore

3.1.5 Process Modeling (DFD)



Figure 3.5: Level 0 DFD of Online Bookstore

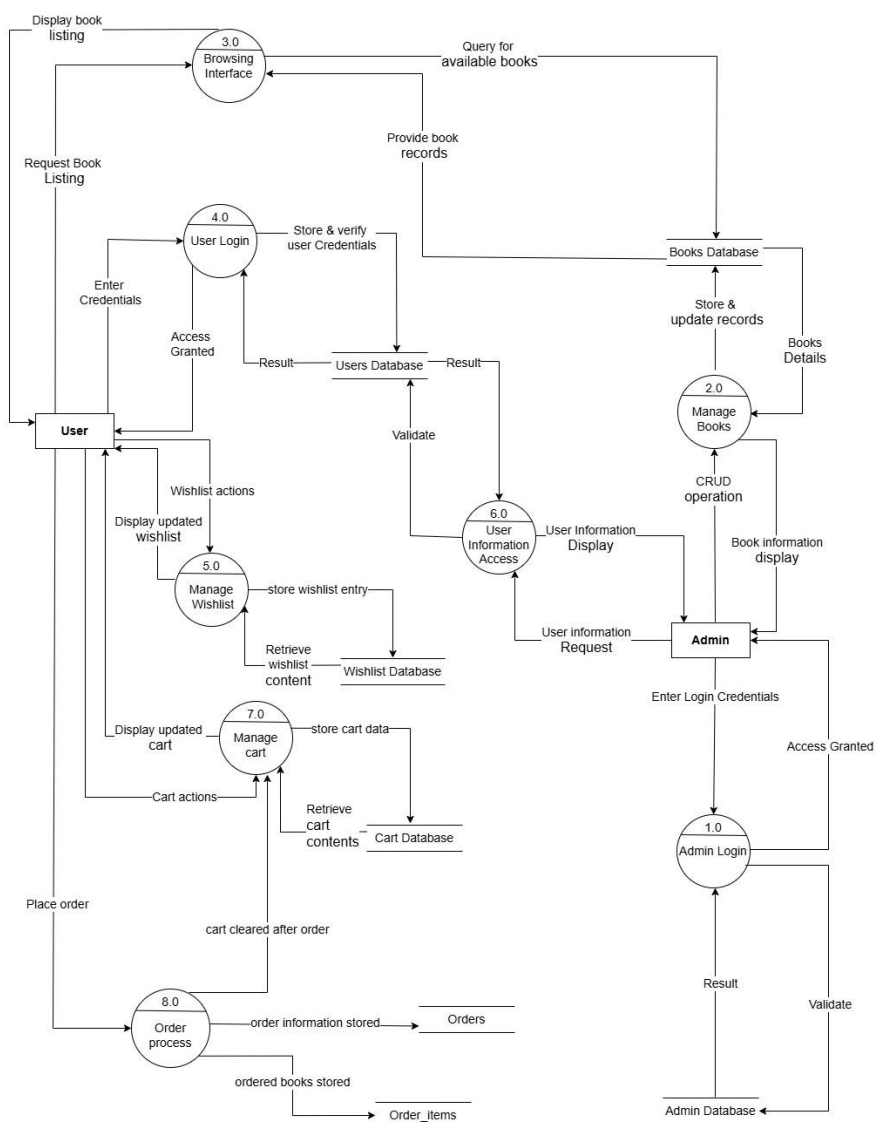


Figure 3.6: Level 1 DFD of Online Bookstore

3.2 System Design

3.2.1 Architectural Design

Components:

- Frontend Layer: HTML5 for structure, CSS3 for styling, and JavaScript for basic interactivity.
- Backend Layer: PHP 8.x for server-side processing, session management, and handling business logic such as job posting, applications, and user roles.
- Database Layer: MySQL for data storage with a normalized relational schema including users, seekers, employers, jobs, and applications.
- File Storage: Local /uploads/resumes/ directory for storing job seeker resume files in PDF format.
- Authentication: Session-based authentication using `password_hash()` and `password_verify()` for secure login and user management.

3.2.2 Database Schema Design

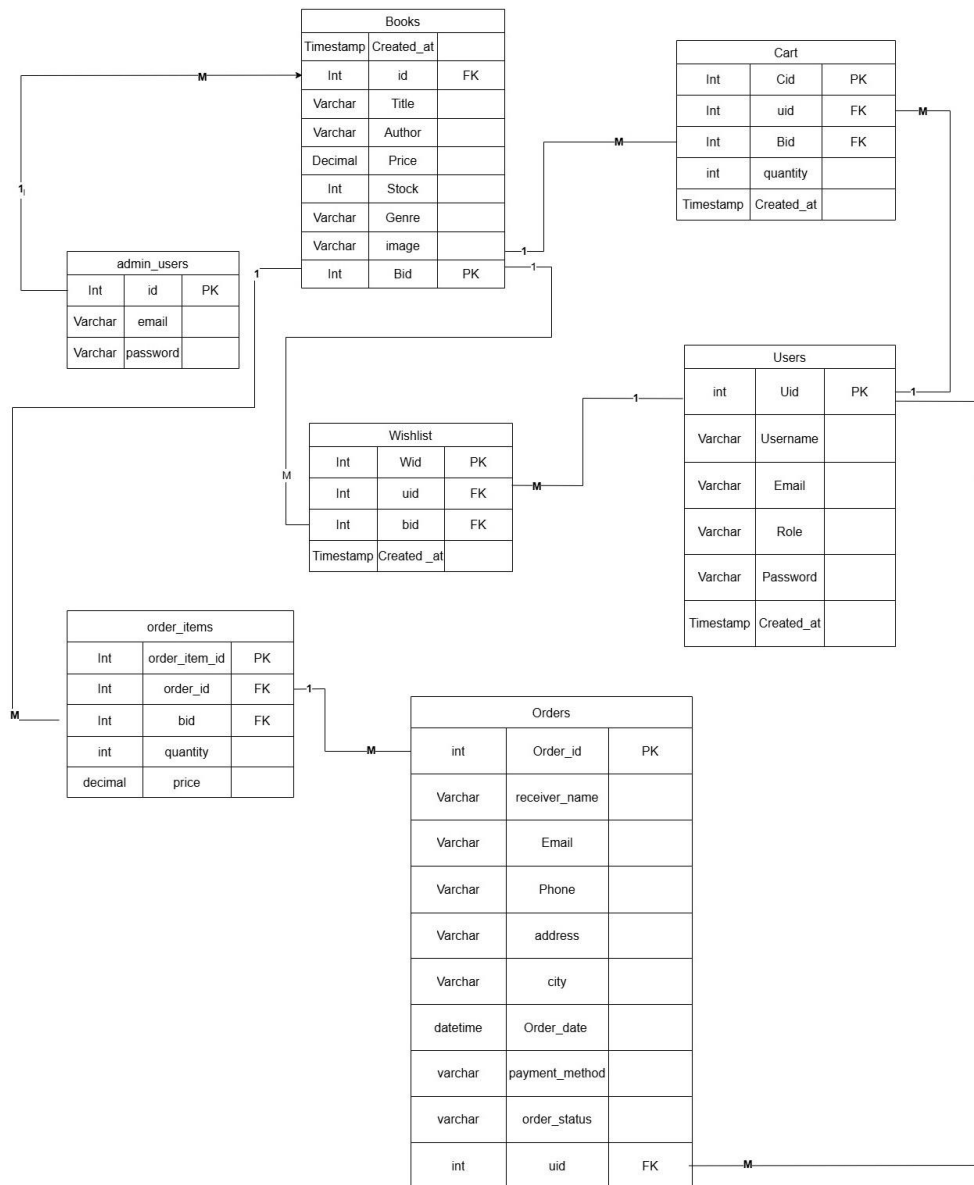


Figure 3.7: Database Schema for Online Bookstore

3.2.3 Interface Design

The user interface follows modern web design principles with clean layouts, intuitive navigation, and responsive components. Each module has a distinct color scheme and layout optimized for its user group. The design emphasizes ease of use with clear call-to-action buttons, proper visual hierarchy, and helpful feedback messages.

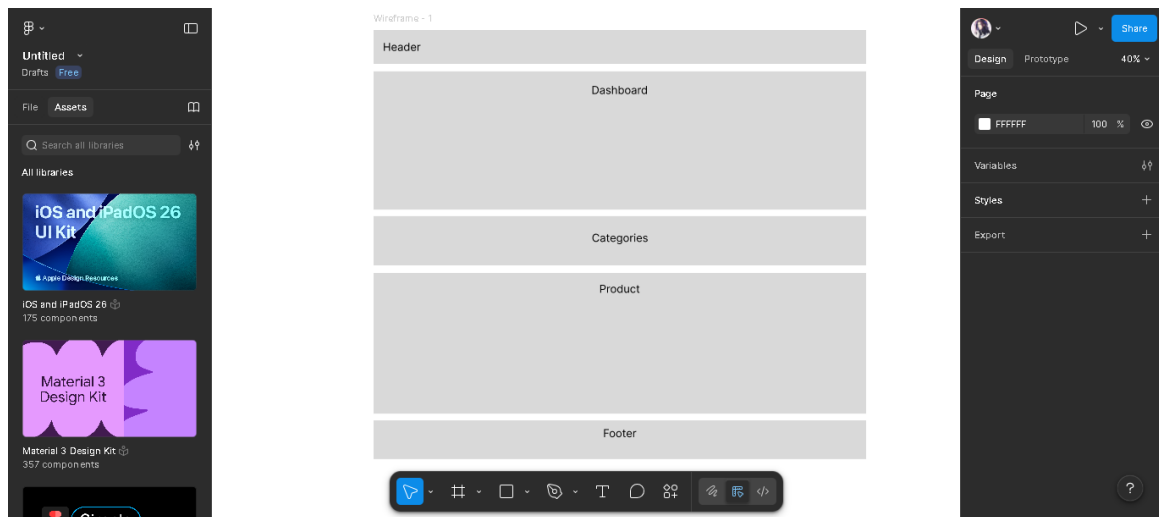


Figure 3.8: Figma Design for User Dashboard

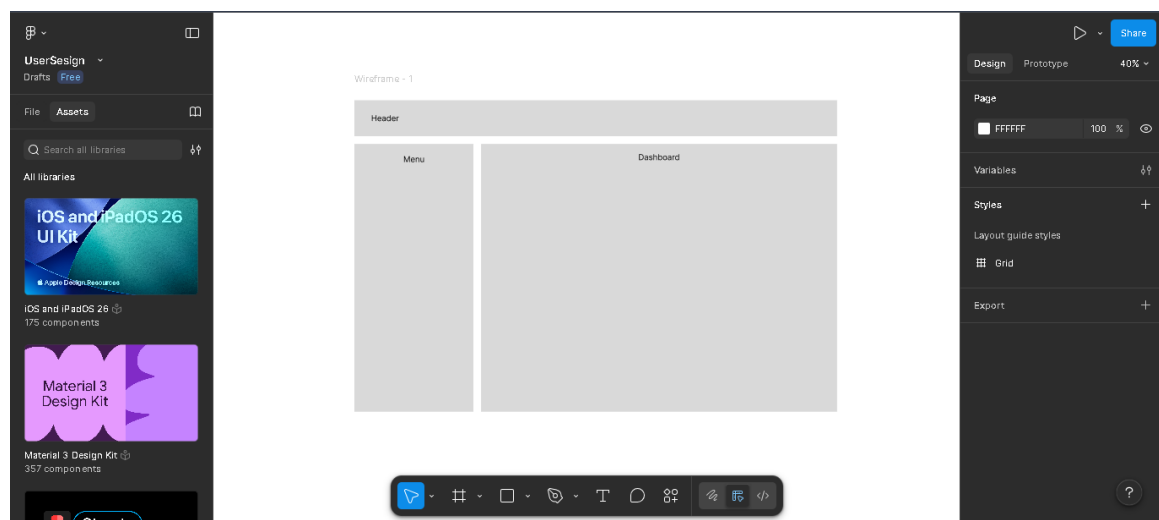


Figure 3.9: Figma Design for Admin Dashboard

Chapter 4: Implementation and Testing

4.1 Implementation

4.1.1 Tools Used

Development Tools:

- IDE: Visual Studio Code with PHP, HTML, CSS, and JavaScript extensions.
- Local Server: XAMPP (Apache, MySQL, PHP 8.x).
- Database Management: phpMyAdmin for managing MySQL database.
- Version Control: Git for tracking code changes and maintaining versions.
- Email Service: Gmail SMTP (via PHPMailer) for sending email notifications.

Programming Languages and Technologies:

- Backend: PHP 8.x (server-side logic, session management)
- Frontend: HTML5 (structure), CSS3 (styling), JavaScript (interactivity)
- Database: MySQL 8.0 (data storage and management)

4.1.2 Implementation Details

1. Admin Module

The Admin module acts as the central authority for managing the overall job portal system, ensuring that users and job postings are properly maintained and monitored.

- **Job Management (CRUD Operations):** Admins can view, edit, and delete any job posted on the platform. Changes are immediately reflected across the system.
- **User Oversight:** Provides access to view all registered users (job seekers and employers) for monitoring purposes.
- **Platform Control:** Ensures that inappropriate or outdated job listings are removed to maintain system quality.
- **Access Control:** Admin privileges allow secure handling of sensitive operations, ensuring only authorized personnel can manage users and jobs.

2. Employer Module

The Employer module provides tools for employers to manage job postings and interact with applicants efficiently.

- **Job Posting (CRUD Operations):** Employers can create, update, delete, and view their job listings.
- **Applicant Management:** Employers can view applicants for each job along with their details and resumes.
- **Application Status Update:** Employers can accept or reject applications, which updates the system and triggers email notifications.
- **Profile Management:** Employers can update company-related information and manage their account details.

3. Job Seeker Module

The Job Seeker module provides an interface for users to search and apply for jobs and manage their profiles.

- **Job Browsing:** Job seekers can search and filter jobs based on keywords and categories.
- **Job Application:** Job seekers can apply for jobs using their profile and uploaded resume.
- **Application Tracking:** Job seekers can view the status of their applications (pending, accepted, rejected).
- **Profile Management:** Job seekers can update personal details, skills, and upload resumes for job applications.

Key Implementation Features:

- **Job Posting Workflow:** Employers can create and publish job listings that are instantly visible on the job board.
- **Job Application System:** Job seekers can apply using their profile and resume, with checks for duplicate applications and deadlines.
- **Application Tracking:** Job seekers can view application statuses (pending, accepted, rejected) for transparency.
- **Email Notifications:** Automated emails are sent to job seekers when application status is updated.
- **Security Measures:** Password hashing and session-based authentication ensure secure user access.

4.2 Testing

The system underwent comprehensive testing, including unit testing of individual modules, integration testing of module interactions, and system testing of complete workflows. Testing focused on functional correctness, security validation, and user experience quality.

Testing Approach:

- Unit Testing: Individual functions, forms and database queries tested for correctness.
- Integration Testing: Module interactions verified (e.g., job application affecting employer's applicants list)
- System Testing: End-to-end workflows tested from user registration to job application and status updates.
- Security Testing: Password hashing, session management.
- Usability Testing: Interface tested with potential users for ease of navigation and clarity.

4.2.1 Test Cases for Unit Testing

These tests focus on individual module functionalities like user authentication, job posting, and application management.

Table 4.2.1.1 User Authentication and Registration

Test case Id	Test scenario	Test steps	Test data	Expected result	Actual result	Pass /Fail
TC01	User Registration	1.Navigate to signup 2.Enter user details 3.Click Submit	Username: Rohit Email: rohit@example.com Password: Rohit@123 Role: Seeker	System creates user record and redirects to respective dashboard	User registered and redirected	Pass

TC02	User Login (Authorized)	1.Go to login page 2.Enter valid credentials 3. Click Login	Email: rohit@example.com Password: Rohit@123	User logged in and redirected to dashboard based on role	User logged in successfully	Pass
TC03	User Login (Unauthorized)	1.Go to login page 2.Enter invalid credentials 3. Click Login	Email: rohit@example.com Password: wrong pass	Error message shown: "Invalid email or password"	Error message displayed	Pass

Table 4.2.1.2 Job, Application & Profile Management

Test case Id	Test scenario	Test steps	Test data	Expected result	Actual result	Pass /Fail
TC04	Job Posting (Employer)	1. Employer logs in 2. Fills job posting form 3. Click Submit	Job Title: "Web Developer" Location: Kathmandu Salary: 50,000 Deadline: 2026-04-15 Category: IT	Job appears on public job board	Job added successfully	Pass

TC05	Job Application (Seeker)	1. Job seeker logs in 2. Opens job detail 3. Click "Apply"	Job ID: 101	Application saved in database, status set to 'pending'	Application submitted successfully	Pass
TC06	Application Status Update (Employer)	1. Employer logs in 2. Views applicants 3. Click Accept/Reject	Application ID: 201	Application status updated in database Email notification sent	Status updated successfully	Pass
TC07	Profile Update (Seeker)	1. Job seeker logs in 2. Opens profile page 3. Updates skills/resume 4. Click Save	Skills: PHP, MySQL Resume: resume.pdf	Profile details updated, resume uploaded to /uploads/resumes	Profile updated successfully	Pass
TC08	Job Deletion (Admin)	1. Admin logs in 2. Selects job 3. Click Delete	Job ID: 102	Job removed from database along with applications	Job deleted successfully	Pass

4.2.2 Test Cases for System Testing

System testing validates the end-to-end workflow of the Online Job Portal, ensuring all modules interact correctly and behave as expected.

Table 4.2.2.1 Test Cases for System Testing

Test case Id	Test scenario	Test steps	Test data	Expected result	Actual result	Pass /Fail
ST01	User Registration & Login	1.Navigate to signup 2.Enter details 3.Submit 4.Login	User Name: Ram Email: ram@example.com Password: ram@123	User account created, login successful, dashboard shown	User registered and logged in	Pass
ST02	Employer Adds Job	1.Employer logs in 2. Navigate to Add Job 3. Enter job details 4. Submit	Job Title: “Web Developer” Location: Kathmandu Salary: 50,000 Deadline: 2026-04-15 Category: IT	Job added to system, visible to job seekers	Job visible on job board	Pass
ST03	Employer Updates Job	1.Employer logs in 2. Navigate to My Jobs 3. Edit job details 4. Submit	Job ID: 101 Salary: 55,000	Job details updated successfully, reflected on job board	Job updated successfully	Pass

ST04	Job Seeker Applies	1. Job seeker logs in 2. Open job details 3. Click Apply	Job ID: 101 Resume: resume.pdf	Application saved in database, status set to 'pending'	Application submitted successfully	Pass
ST05	Employer Updates Application Status	1. Employer logs in 2. View applicants 3. Click Accept/Reject	Application ID: 201	Application status updated, email notification sent	Status updated successfully	Pass
ST06	Admin Deletes Job	1. Admin logs in 2. Selects job 3. Click Delete	Job ID: 102	Job removed from system, no longer visible to job seekers	Job deleted successfully	Pass

4.2.3 Test Cases for Integration Testing

Integration testing checks that all modules work together correctly, ensuring smooth interactions between Admin, Employers, and Job Seekers and maintaining data consistency across the system.

Table 4.2.3.1 Test Cases for Integration Testing

Test case Id	Test scenario	Test steps	Test data	Expected result	Actual result	Pass /Fail
IT01	User Registration	1. Register new user 2. Login with same credentials	Name: Ramesh Email: ramesh@example.com Password: ramesh@123	User created, login successful, dashboard displayed	User registered and logged in	Pass
IT02	Employer Adds Job & Seeker Views	1. Employer logs in 2. Adds new job 3. Job seeker logs in and browses jobs	Job Title: Web Developer Location: Kathmandu Salary: 50,000 Deadline: 2026-04-15 Category: IT	Job added by employer appears on job board for seekers	Job visible on job board	Pass
IT03	Job Seeker Applies	1. Job seeker logs in 2. Opens job details 3. Click Apply	Job ID: 101 Resume: resume.pdf	Application saved in database, status set to 'pending'	Application submitted successfully	Pass

IT04	Employer Updates Application Status	1. Employer logs in 2. Views applicants 3. Click Accept/Reject	Application ID: 201	Application status updated in database, email notification sent	Status updated successfully	Pass
IT05	Admin Deletes Job	1. Admin logs in 2. Selects job 3. Click Delete	Job ID: 102	Job removed from system, no longer visible to seekers	Job deleted successfully	Pass

Chapter 5: Conclusion and Future Recommendations

5.1 Lesson Learnt / Outcome

The Online Job Portal successfully achieves its main objective of providing a streamlined platform for connecting job seekers with employers. By focusing on core functionalities, the system allows administrators to efficiently manage users and job postings, while offering employers and job seekers intuitive interfaces for job management and applications.

Key achievements include:

- Fully functional platform with distinct Admin, Employer, and Job Seeker modules.
- Comprehensive CRUD operations for employers to manage job postings.
- Application management for job seekers, including applying, tracking, and updating resumes.
- Real-time synchronization between administrative actions and user-facing pages.
- Clean, maintainable code structure with organized external CSS and JavaScript.
- Secure authentication and safe handling of user data.
- Automated email notifications for application status updates.

5.2 Conclusion

This project demonstrates that a well-structured digital platform can effectively facilitate job recruitment in an online environment. The system balances administrative oversight with user-focused functionality, including job posting, application submission, profile management, and status tracking.

The modular architecture and organized code ensure maintainability and allow for future enhancements, such as adding advanced search, filtering, or recommendation features. Administrators benefit from efficient monitoring and management of users and job postings, employers can manage vacancies and applicant responses easily, and job seekers enjoy a personalized experience with tracking and resume management.

Although some advanced features like AI-based job matching or payment processing for premium job postings are not included, the system provides a solid foundation for a secure, user-friendly, and functional online job portal. This project validates the practicality of integrating administrative control with intuitive user interfaces in a web-based recruitment platform.

5.3 Future Recommendations

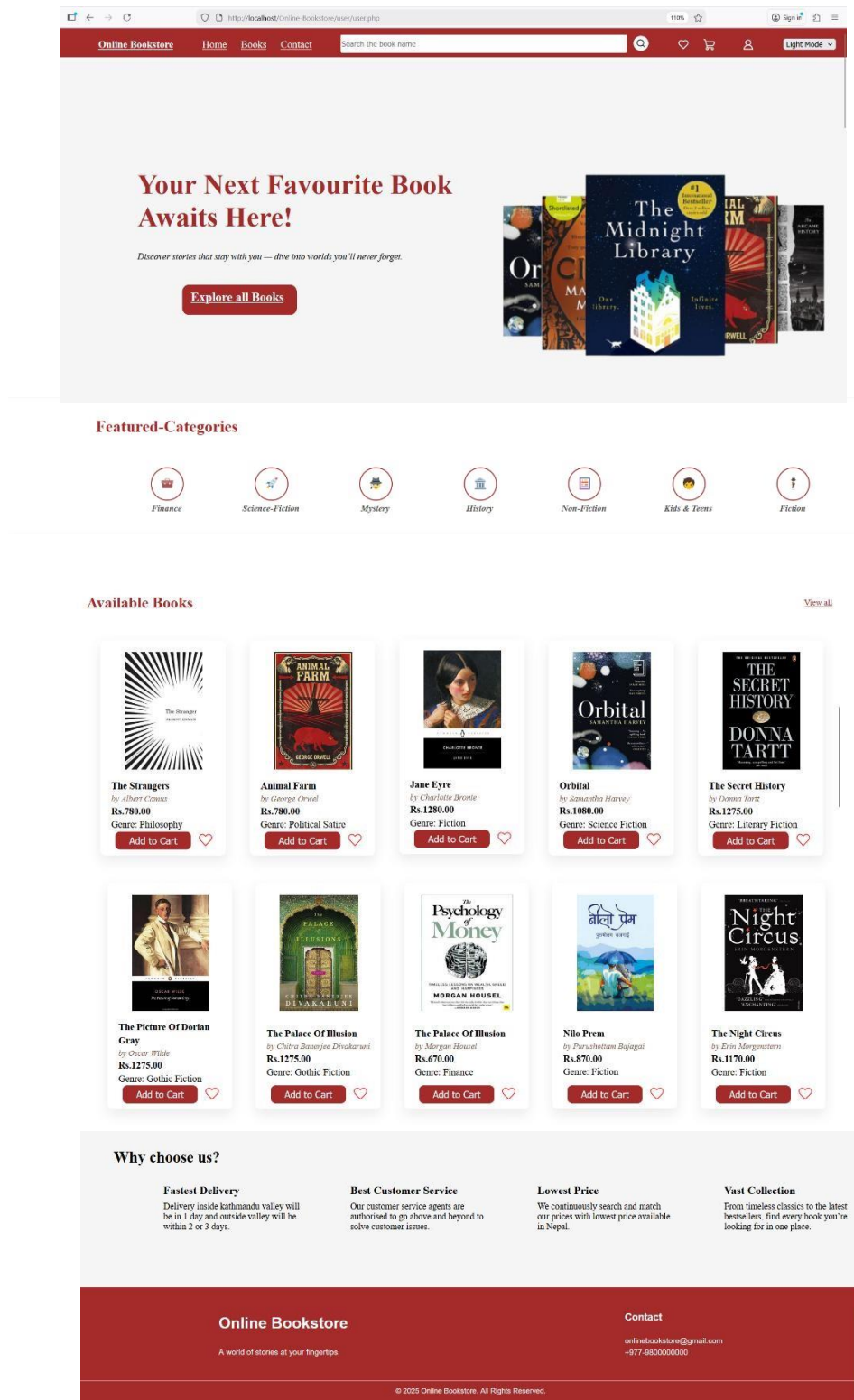
To further improve the functionality and user experience of the Online Job Portal, the following enhancements are recommended for future development:

- **Employer Analytics and Reporting:** Provide employers with insights on job post views, application trends, and candidate engagement to support informed decision-making.
- **Resume and Profile Recommendations:** Introduce algorithms to suggest suitable jobs to seekers based on their skills, experience, and application history.
- **Enhanced User Features:** Add features like company reviews, job ratings, and applicant feedback to increase engagement and transparency.
- **Responsive and Accessible Design:** Further optimize the portal for mobile devices and accessibility standards to reach a wider audience.
- **Notification Enhancements:** Expand notifications to include reminders for application deadlines, interview schedules, or status updates.
- **Support for Multiple Job Types:** Include part-time, freelance, and internship options alongside full-time positions to cater to diverse job seekers.
- **Interview Scheduling System:** Allow employers to schedule interviews through the portal and notify applicants automatically..

REFERENCES

- [1] A. Sundararajan, “The Economics of Digital Platforms: Understanding Online Marketplaces”, MIT Press, 2016.
- [2] R. Kaur, S. Kaur, and M. Singh, “A Review of Personalized Job Recommendation Systems”, International Journal of Computer Applications, vol. 182, no. 38, pp. 15–22, 2018.
- [3] LinkedIn, “How LinkedIn Matches Candidates to Jobs”, [Online]. Available: <https://www.linkedin.com/help/linkedin/answer/61615>. [Accessed: Mar. 25, 2026].
- [4] Indeed, “Job Search & Career Platform Features”, [Online]. Available: <https://www.indeed.com/hire/c/info>. [Accessed: Mar. 25, 2026].
- [5] R. W. Sebesta, Programming the World Wide Web, 9th ed., Boston, MA: Pearson, 2019.
- [6] PHPMailer, “PHPMailer: Sending Email via SMTP”, [Online]. Available: <https://github.com/PHPMailer/PHPMailer>. [Accessed: Mar. 25, 2026].

APPENDICES



Listing A: User Dashboard

Online-Bookstore

Home

Books

Users

Add Books

Update Books

Pending Order

Completed Order

Log Out

Home

Admin Dashboard

Total Books

15

Total Users

6

Pending Orders:

5

Completed Orders:

9

Pending Order List

View all

S.N	Receiver	Phone	Email	Address	City	Date	Items	Action	Total Price
1	Manisha Magar	9876555646	manisha45@gmail.com	Thecho	Lalitpur	2026-03-20 13:43:42	The Alchemist	Pending	670.00
2	Rachana Gautam	9765467439	rachanag@gmail.com	Takhel	Lalitpur	2026-03-20 13:43:04	Orbital	Pending	1080.00
3	Krish Maharjan	9876547890	krishmhzn@gmail.com	Dhapakhel	lalitpur	2026-03-20 13:42:36	The Palace Of Illusion	Pending	1275.00
4	Sakshyam Shrestha	9808765432	sakshyamstha12@gmail.com	lagankhel	lalitpur	2026-03-20 13:41:57	Crime and Punishment, Lords of Crypto Crime	Pending	2468.00

Listing B: Admin Dashboard

